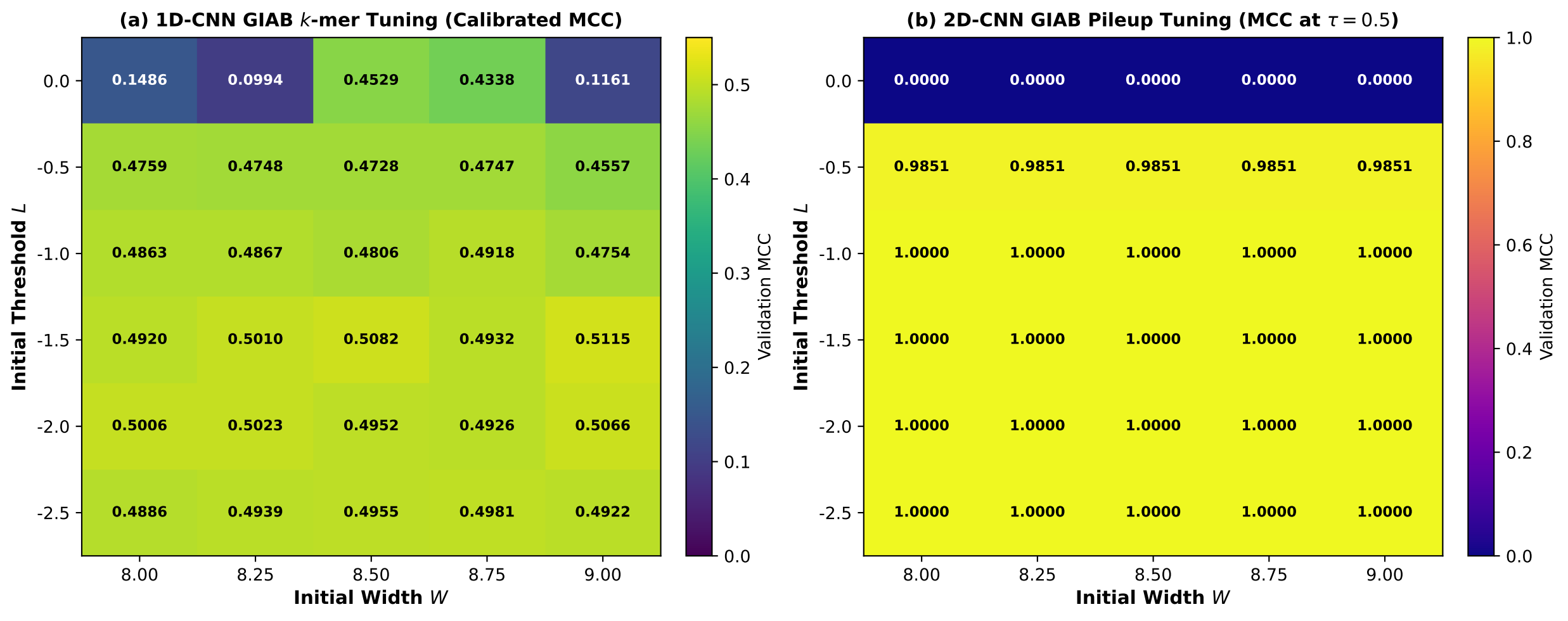
(a) 1D-CNN GIAB k-mer Tuning (Calibrated MCC)
Initial Threshold L
Initial Width W
Validation MCC
0.0
-0.5
-1.0
-1.5
-2.0
-2.5
8.00
8.25
8.50
8.75
9.00
0.1486 0.0994 0.4529 0.4338 0.1161
0.4759 0.4748 0.4728 0.4747 0.4557
0.4863 0.4867 0.4806 0.4918 0.4754
0.4920 0.5010 0.5082 0.4932 0.5115
0.5006 0.5023 0.4952 0.4926 0.5066
0.4886 0.4939 0.4955 0.4981 0.4922
0.5
0.4
0.3
0.2
0.1
0.0
(b) 2D-CNN GIAB Pileup Tuning (MCC at τ = 0.5)
Initial Threshold L
Initial Width W
Validation MCC
0.0000 0.0000 0.0000 0.0000 0.0000
0.9851 0.9851 0.9851 0.9851 0.9851
1.0000 1.0000 1.0000 1.0000 1.0000
1.0000 1.0000 1.0000 1.0000 1.0000
1.0000 1.0000 1.0000 1.0000 1.0000
1.0000 1.0000 1.0000 1.0000 1.0000
1.0
0.8
0.6
0.4
0.2
0.0